Mac PySide6 Fix — “cocoa platform plugin not found”

Status: RESOLVED — July 2026
Root cause identified and permanently fixed.

The Error

```
qt.qpa.plugin: Could not find the Qt platform plugin "cocoa" in
"…/.venv/lib/python3.12/site-packages/PySide6/Qt/plugins/platforms"
Abort trap: 6
```

Root Cause

The project was located in `~/Documents/project_atlas/` which is synced by **iCloud Drive**.

iCloud continuously monitors `~/Documents/` and treats any file modification as a potential sync conflict — including the `.dylib` files that pip writes when installing or updating PySide6. Rather than overwrite either copy, iCloud keeps both, renaming the older one to `libqcocoa 2.dylib`, `libqcocoa 3.dylib` etc. It also strips the execute permissions on these copies down to `700`.

Qt cannot load a platform plugin if the filename is wrong or the permissions are broken. The result was `libqcocoa.dylib` periodically getting clobbered or replaced, causing Atlas to crash on launch.

Evidence of the problem — duplicated files visible via `ls -la`:

```
-rwx------   libqcocoa 2.dylib    ← iCloud conflict copy
-rwx------   libqcocoa 3.dylib    ← iCloud conflict copy
-rwxr-xr-x   libqcocoa.dylib      ← real file, worked until next iCloud sync
```

Force-reinstalling PySide6 restored the real file temporarily, but iCloud would break it again on the next sync cycle. This is why the error kept recurring even after successful reinstalls.

Permanent Fix Applied (July 2026)

The project was moved out of `~/Documents/` entirely, to the home folder:

```
mv ~/Documents/project_atlas ~/project_atlas
```

iCloud Drive only syncs `~/Desktop`, `~/Documents`, and `~/Downloads`. The home folder itself (`~/`) is not synced. Moving the project one level up puts it completely outside iCloud’s reach — permanently, with no ongoing maintenance required.

After the move the `.venv` had to be rebuilt because Python venv symlinks are hardcoded to the path they were created at:

```
cd ~/project_atlas
deactivate
rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

Atlas then launched cleanly and the platform plugins have remained intact.

Current Project Location

What	Where
Project root	`~/project_atlas/`
Virtual environment	`~/project_atlas/.venv/`
Launch script	`~/project_atlas/run_atlas.sh`
Database	`~/project_atlas/data/atlas.db`
iCloud exposure	**None**

If This Error Ever Recurs

It shouldn't — but if it does, first check whether the project folder has somehow ended up back inside `~/Documents/`, `~/Desktop/`, or `~/Downloads/`. That's the only way iCloud can touch it again.

If the project location is correct and it still fails, rebuild the venv:

```
cd ~/project_atlas
deactivate
rm -rf .venv
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
./run_atlas.sh
```